

Theo Blauberger

SENIOR ANALYTICS ENGINEER | M.SC. ECONOMICS | M.SC. DATA SCIENCE (ONGOING)

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PROFILE

Senior Analytics Engineer with an advanced degree in Economics and extensive experience in econometrics, machine learning, and experimentation. Currently completing a Master's in Data Science at the University of Helsinki. Proficient in R, Python, SQL, and GPU computing. Experienced in building production ML models, interactive data applications, and internal analytics tooling.

EDUCATION

Master's Program in Data Science

University of Helsinki

09/2021 -

- **Major:** Data Science
- Specializing in machine learning methods with emphasis on GPU-accelerated computing.
- **Thesis:** Applying tabular foundation models to the detection of fraud patterns in high-dimensional transaction data. Work in progress; **current draft**.

Master's Program in Economics

University of Helsinki

09/2020 - 12/2022

- **Major:** Economics
- **Minors:** Statistics, Mathematics, Computer Science
- **Master's Thesis:** Impact of Policy Shocks in the EU Emissions Trading System on Finland.

Exchange Program

Fudan University, Shanghai

09/2018 - 06/2019

Studied Chinese language, politics, and economics, with focus on East Asian economic development.

Language Course

University of Vienna

09/2015 - 12/2015

Intensive German language studies.

TECHNICAL SKILLS

Languages: R, Python, SQL, Julia, C++

Methods: causal inference, RCT analysis, time-series econometrics

ML/AI: Torch, tidymodels, scikit-learn, Claude Agent SDK, Anthropic Python SDK, Chatlas

WORK EXPERIENCE

Senior Analytics Engineer

Nordea

07/2025 -

- Build, maintain, and monitor machine learning models that deliver real-time fraud detection for card and account-to-account transactions.
- Lead analytics initiatives with increased autonomy in model development and deployment decisions.
- Maintain an internal R package for data analysis and visualization, significantly improving team productivity.

Data Analyst

Nordea

07/2023 - 07/2025

- Collaborated within a team responsible for building and monitoring real-time fraud detection models.
- Developed internal R packages, Shiny apps, and automated reports. Established a Data & Analytics community within the department.

Data Analyst

VATT Institute for Economic Research

01/2023 - 06/2023

- Conducted analyses and co-authored research reports on electricity market data, focusing on **consumer** and **company** responses to price shocks.
- Designed and developed a **Shiny-based dashboard** for interactive data visualization.

Research Assistant

VATT Institute for Economic Research

08/2022 - 01/2023

- Supported research projects with data preparation, analysis, and visualization tasks.

Project Worker

City of Tampere

05/2022 - 08/2022

- Analysed a randomised controlled experiment (~5,000 users) testing whether information nudges could shift mobility behaviour, as part of the Keli project (Kestävemmän liikkumisen kehittäminen hiilijalanjälkilaskurin avulla).
- Project co-funded by the Ministry of the Environment. Results published in a **working paper**.

Intern

Embassy of Finland in Vienna

05/2021 - 08/2021

- Monitored and reported on Austrian economic developments to inform Finnish Government policy decisions.
- Attended and reported on meetings with UN organizations and local politicians.

Data: dplyr, DuckDB, pandas, Arrow, Spark, Hive, Impala, Snowflake
GPU: CUDA Python, RAPIDS cuDF
Engineering: Docker, GitHub Actions, pytest, testthat
Web: Shiny, Quarto, PowerBI, REST APIs

CERTIFICATIONS

- **NVIDIA:** Fundamentals of Deep Learning
- **NVIDIA:** Fundamentals of Accelerated Computing with CUDA Python

PROJECTS

Running Coach - Self-Hosted Training Planner

- Building a self-hosted running coach that ingests Apple Health data into a canonical activity and vitals schema, maintains fitness and injury-load ledgers, and generates weekly plans from a deterministic guardrail engine.
- A Claude Agent SDK reviewer proposes plan changes that are validated against the guardrails before application. Python, FastAPI, SQLite, Docker; deployed on a home server behind Tailscale with pytest coverage across the engine.

cuplyr - GPU-Accelerated dplyr

- Developing an R package that enables standard dplyr code to execute on GPU hardware through a RAPIDS cuDF backend.
- Implements lazy evaluation with automatic query optimizations. Benchmarks show 40-77x speedups over dplyr on GPU-resident data; parity end-to-end including transfer.
- [GitHub](#)

digitraffic - Fintraffic API Client

- Building an R package for Finland's Digitraffic road traffic sensor time series: 450+ roadside sensors, per-vehicle records, real-time and historical.
- Features tidyverse-native output, spatial filtering, built-in caching, and rate limiting. Tested with GitHub Actions; to be submitted to CRAN.
- [GitHub](#)

bracketeer - Tournament Management Framework

- Developed an R package for modeling and executing tournament competitions with support for multiple formats including round-robin, Swiss system, and elimination brackets.
- Features a pipe-first API design for defining reusable tournament blueprints with automatic stage materialization and flexible result entry.
- [GitHub](#)

Resepor - Recipe Assistant

- [Python Shiny app](#) for interactive recipe creation on the Claude API via Chatlas.

POSITIONS OF RESPONSIBILITY

Vote Counter - UNIDO

Authorized by the Western countries group to serve as vote counter in the Secretary-General Election of the United Nations Industrial Development Organization.

Board Member - Economics Students' Association

Served on the board contributing to strategic planning and student activities.